1. "Praktika Status Overview" Widget

- **What it Shows:** Progress bars with "Allocated X%".
- **The Flaw:** This implies a real-time, incremental process where assignments are made one by one. Our system is a **batch process**. The status will be 0% before the algorithm runs and 100% after. A progress bar is misleading.
- **The Correction:** The goal of this widget is excellent, but it should display the **final state**, not the progress. It should answer the question: "For each internship type, did we successfully place everyone?"
- **What it *Should* Show:**
    - **Demand:** The total number of slots required for each praktikum.
    - **Assigned:** The total number of slots filled.
    - **Unassigned:** The difference. This should always be **0** in a successful run.
- **API Need:** An endpoint that returns a list of objects, one for each practicum type, with these counts.

Note: Instead of a progress bar that implies a process, we should use a simple, clear text-based display or a stacked bar.

Option A (Simple Text - Most Clear):

Praktikum	Demand	Assigned	Unassigned
PDP I	50	50	**0**
PDP II	45	45	**0**
SFP	60	58	**2**

ZSP	55	55	**0**

We can also:

For each internship type, you could have a bar that is visually divided.

- **SFP:** [////////////////////|//]
    - [Green Section] represents the 58 assigned.
    - [Red Section] represents the 2 unassigned.

The API response for this widget will be as follows:

"assignment_status":

[

{

"practicum_type": "SFP",

"demand_slots": 60, // This is the "Demand"

"assigned_slots": 58, // This is the "Assigned"

"unassigned_slots": 2 // This is the "Unassigned"

},

// ... other practicum types

]

2. "Current Year's Budget" Widget

Verdict: No flaws

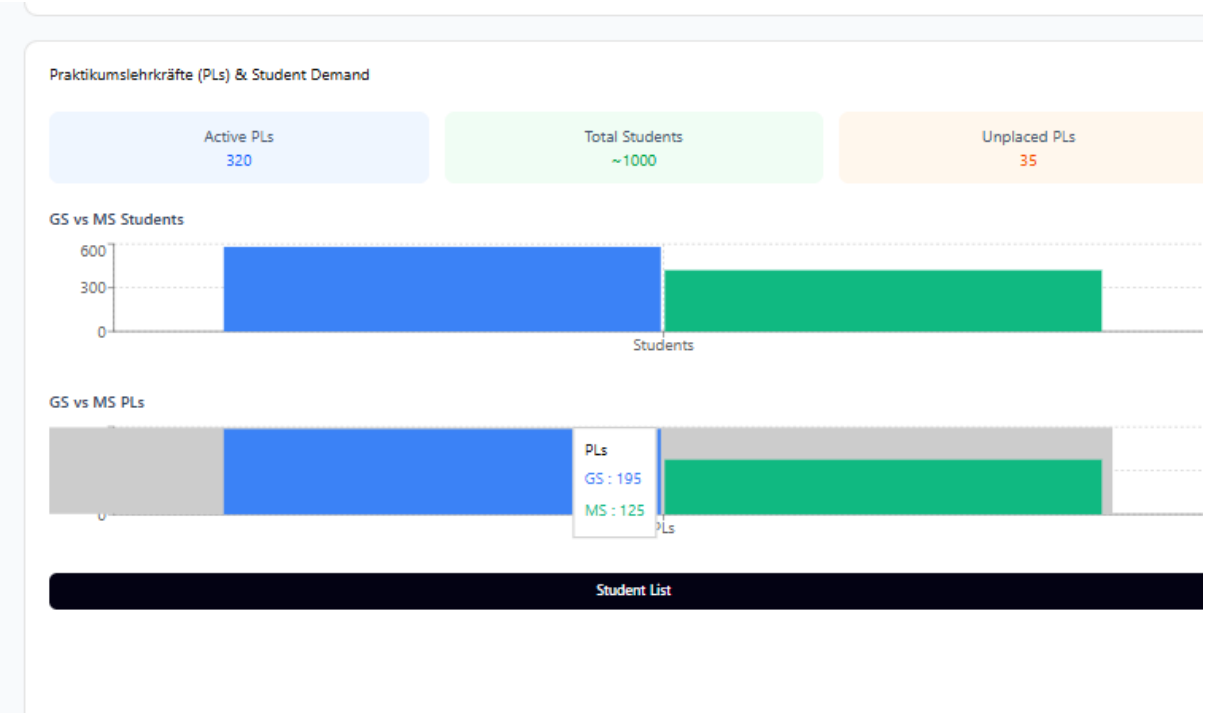
What It Shows (The Business Logic):

This widget provides a complete overview of the most valuable resource the Internship Office has: Credited Hours (Anrechnungsstunden).

- **"Total Budget: 210 Anrechnungsstd."**
  - **Real-World Meaning:** This is the total number of credited hours allocated by the Ministry of Education (`Kultusministerium` or KM) for the entire school year. This number is a hard limit.
  - **Purpose:** It sets the scale for the entire planning operation.
- **"Distributed: 169 GS-PLs, 41 MS-PLs"**
  - **Real-World Meaning:** This shows how the total budget has been allocated to the available mentors. It is the sum of the `anrechnungsstunden` for every active mentor in the system, broken down by their program (`LA = GS` or `MS`).
  - **Purpose:** It shows the administrator the total "spending power" they have. In this case, `169 + 41 = 210`, which perfectly matches the total budget.
- "Remaining: 5 Anrechnungsstd."
  - Real-World Meaning: This is the most critical metric. It's the Total Budget minus the Distributed budget. It should be zero after a successful allocation. A positive number means there are unallocated hours (a buffer). A negative number is a major red flag, meaning they have promised more hours to mentors than they officially have.
- **The Pie Chart (GS-PLs 80% / MS-PLs 20%)**
  - **Real-World Meaning:** This is a visual representation of the "Distributed" numbers (`169 / 210` is approx. 80%).
  - **Purpose:** It gives the administrator a quick, intuitive sense of how their mentor resources are split.

API Response:


```
{
"budget_summary": {
 "total_budget": 210,
 "distributed_gs": 169.0,
 "distributed_ms": 41.0,
 "total_distributed": 210.0,
 "remaining_budget": 0.0
 }
 }
```



3. "Praktikumslehrkräfte (PLs) & Student Demand" Widget

The Flaw: The metric "Unplaced PLs" is confusing and not meaningful. A PL is a resource; they are either *used* or *unused*. The truly critical metric, which is shown in your "System Alerts" widget, is "Unplaced Students."

Widget Design & Logic:

Widget Title: Praktikumslehrkräfte (PL) & Student Demand

Top Row: Key Performance Indicators (KPIs)

This row should show three main numbers:

1. **Active PLs**
    - **Meaning:** The total number of mentors who are available for assignment in the current planning year (i.e., not ruhend, on Elternzeit, etc.).
    - **Calculation:** PraktikumsLehrkraft.objects.filter(is_active=True).count()
    - **Example Value:** 210
2. **Total Students**
    - **Meaning:** The total number of students in the system who require at least one internship placement for the current planning cycle.
3. **Unplaced Students** (This is the corrected metric)
    - **Meaning:** The number of students who **could not be assigned** an internship slot after the main algorithm has run.
    - **Example Value (Ideal):** 0

- **Example Value (Problem):** 12 (This is the number that would trigger the "System Alert").

Bottom Row: Bar Charts for Comparison

These charts break down the PIs by program type (GS vs. MS).

1. **GS vs MS Students Bar Chart:**
   - **Meaning:** A simple bar chart showing the distribution of total unplaced students between the GS and MS programs.
   - **Data Needed:**
     - Count of UNPLACED students where program='GS'.
     - Count of UNPLACED students where program='MS'.
   - **Purpose:** Helps the admin see if there's a problem specific to one program (e.g., "Why are there so many unplaced MS students?").
2. **GS vs MS PLs Bar Chart:**
   - **Meaning:** A simple bar chart showing the distribution of available mentor resources between the GS and MS programs.
   - **Data Needed:**
     - Count of active PLs where program='GS'.
     - Count of active PLs where program='MS'.
   - **Purpose:** Allows the admin to visually compare the supply ratio (e.g., "We have far more GS mentors than MS mentors"). This can help explain why there might be unplaced MS students.

API Response:

{

"entity_counts": {

"total_students": 1023, // Powers the "Total Students" KPI

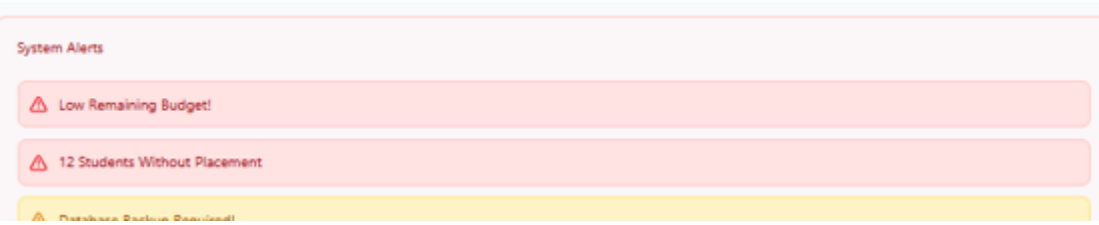
"unplaced_students": 12, // Powers the "Unplaced Students" KPI and the System Alert
"active_pls_total": 210, // Powers the "Active PLs" KPI

"active_pls_gs": 169, // Powers the GS bar in the "GS vs MS PLs" chart

"active_pls_ms": 41, // Powers the MS bar in the "GS vs MS PLs" chart
"unplaced_students_gs": 8, // Data needed for the "GS vs MS Students" chart
"unplaced_students_ms": 4 // Data needed for the "GS vs MS Students" chart

}

}



4. "System Alerts" Widget

- **What it Shows:** Actionable alerts like "Low Remaining Budget!" and "12 Students Without Placement".
- **Verdict: This is the most valuable widget on the page.** It translates raw data into urgent tasks for the administrator. This is excellent.
- **API Need:** No new API is needed. The frontend can generate these alerts based on the data from the other API endpoints (e.g., if (remaining_budget < 10) { show_alert(); }).

For the upcoming deadlines for now Don't include it

The final API response you will get is:


```
{
 "assignment_status": [
  {
   "practicum_type": "PDP I",
   "demand_slots": 50,
   "assigned_slots": 50,
   "unassigned_slots": 0
  },
  {
   "practicum_type": "PDP II",
   "demand_slots": 45,
   "assigned_slots": 45,
   "unassigned_slots": 0
  },
  {
   "practicum_type": "SFP",
```

```
    "demand_slots": 60,
    "assigned_slots": 58,
    "unassigned_slots": 2
  },
  {
    "practicum_type": "ZSP",
    "demand_slots": 55,
    "assigned_slots": 55,
    "unassigned_slots": 0
  }
],
"budget_summary": {
  "total_budget": 210,
  "distributed_gs": 169,
  "distributed_ms": 41,
  "remaining_budget": 5
},
"entity_counts": {
  "total_students": 1023,
  "unplaced_students": 2,
  "active_pls_total": 210,
  "active_pls_gs": 169,
  "active_pls_ms": 41
}
}
```